

Thèmes de dynamique en dimension 1 et 2

Exemples in small dimension

Renormalisation: • Rotation on S^1

• Interval exchange maps

• Homeos of S^1
diffs

• Quadratic family

• Hénon family

$$\begin{aligned} f_\lambda(x) &= \lambda x(1-x) \quad \lambda \in \mathbb{R}_+^* \\ f_\lambda: \mathbb{R} &\rightarrow \mathbb{R} \\ x_0 \in \mathbb{R}, \quad x_{n+1} &= f_\lambda(x_n) \end{aligned}$$

Operator R acting on dynamical sys $\mathbb{R} \rightarrow \mathbb{R}$
look at the dynamics of this operator
→ deduce the dynamics of f_λ

I Rotations of the circle:

$$\alpha \in \mathbb{R} \quad S^1 = \mathbb{R} / \mathbb{Z} \quad R_\alpha: x \mapsto x + \alpha \pmod{\mathbb{Z}}$$

1st goal show the 3 gaps

Thm "3 longueurs" $\forall n \in \mathbb{N} \quad \forall x \in S^1 \quad S^1 \setminus \{R_\alpha^k(x)\}_{0 \leq k \leq n-1}$

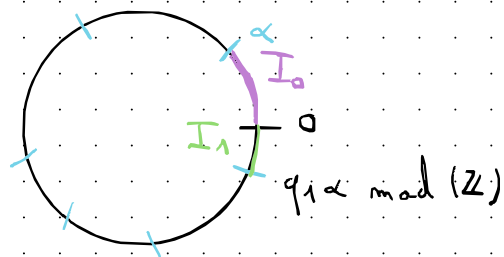
is made of intervals of at most 3 different lengths

Conjectured by Steinhaus Solved in the 50's

Let's analyse the orbits of R_α

Rk: Sufficient to take $x=0$

Suppose that $\alpha \in [0, \frac{1}{2}]$

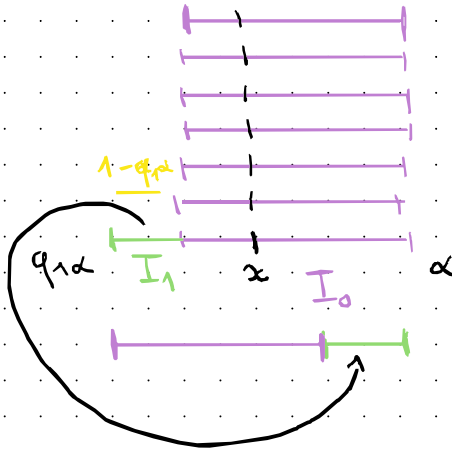
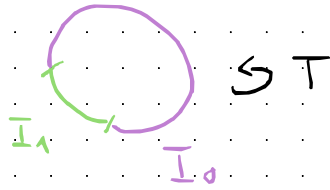


Return map on $I_0 \cup I_1$

$$x \in S^1 \mapsto T(x) = \min \{n \in \mathbb{N}^* \mid R_\alpha^n(x) \in I_0 \cup I_1\}$$

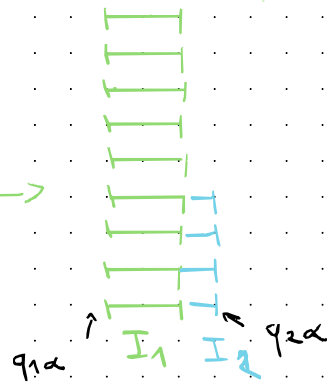
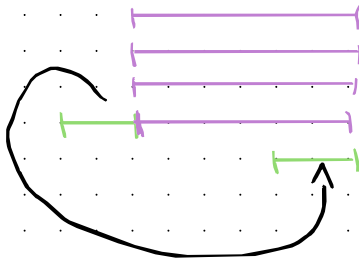
$$T \text{ return map } T(x) = R_\alpha^{T(x)}(x)$$

Lemma: If I glue the ends of $I_0 \cup I_1$
 T is a rotation



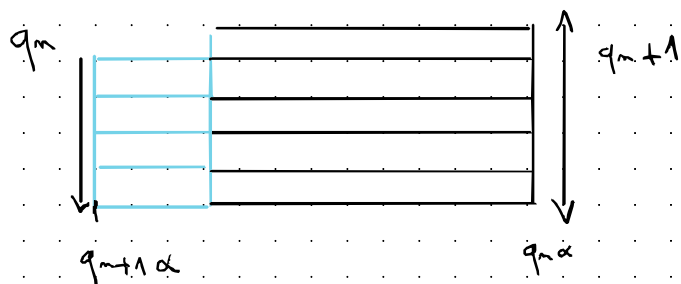
T acts as an exchange of 2 intervals, this is a rotation on $I_0 \cup I_1$ "glued"

$$\text{angle} = \frac{|I_0|}{|I_0| + |I_1|} = \frac{d}{1 + \alpha - q_1\alpha}$$



$$q_{n+1} = \min \{ q > q_n \mid \|q_n \alpha\| > \|q \alpha\| \}$$

$$\text{for } x \in S^1 \quad \|x\| = d(x, \mathbb{Z}) \in [0, \frac{1}{2}]$$

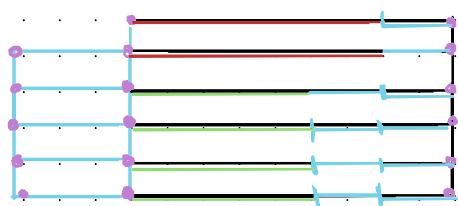


Re: By this construction, $q_n(q_n \alpha - p_n) = (-1)^n p_n$

p_n or $\|q_n \alpha\| = q_n \alpha - p_n$

Proof of the "3 languages" thm:

Take $k \in \mathbb{I} q_n, q_{n+1} \mathbb{I}$



Re: $1 = q_n \times |q_{n+1} \alpha - p_{n+1}| + q_{n+1} |q_n \alpha - p_n|$

$q_{n+1} > q_n$ and $|q_{n+1} \alpha - p_{n+1}| < |q_n \alpha - p_n|$

$\bullet 1 \geq q_{n+1} |q_n \alpha - p_n|$

$\Rightarrow |\alpha - \frac{p_n}{q_n}| \leq \frac{1}{q_{n+1} q_n}$

" $\frac{p_n}{q_n}$ are good approx of α "

Question how to compute p_n and q_n ?

$a_{n+2} = \left\lfloor \frac{|I_n|}{|I_{n+1}|} \right\rfloor = \left\lfloor \frac{\|q_n \alpha\|}{\|q_{n+1} \alpha\|} \right\rfloor$

$\bullet q_{n+1} |q_n \alpha - p_n| > \frac{1}{2}$

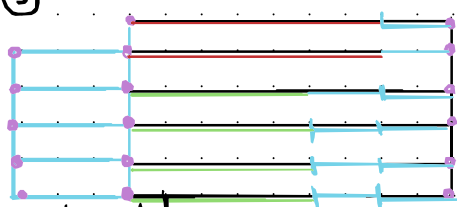
$\Rightarrow |\alpha - \frac{p_n}{q_n}| > \frac{1}{2 q_n q_{n+1}}$

Ex: Proof that $q_{n+2} = a_{n+2} q_{n+1} + q_n$ ①

$p_{n+2} = a_{n+2} p_{n+1} + p_n$ ②

$\|q_{n+2} \alpha\| = -a_{n+2} \|q_{n+1} \alpha\| + \|q_n \alpha\|$

③



$\Rightarrow \|q_n \alpha\| = a_{n+2} \|q_{n+1} \alpha\| + \|q_{n+2} \alpha\|$

① $q_{n+2} = \min \left\{ q > q_{n+1} : \|q \alpha\| > \|q_{n+1} \alpha\| \right\}$

$q_{n+2} = q_n + a_{n+2} q_{n+1}$

② $p_{n+2} (-1)^{n+2} \|q_{n+2} \alpha\| = q_{n+2} \alpha - p_{n+2}$

$p_{n+2} = (q_{n+2} \alpha + (-1)^{n+1} \|q_{n+2} \alpha\|)$

$p_{n+2} = q_n \alpha + a_{n+2} q_{n+1} \alpha + (-1)^{n+1} \|q_{n+2} \alpha\|$

$p_{n+2} = (q_n \alpha + (-1)^n \|q_n \alpha\|) + a_{n+2} q_{n+1} \alpha + a_{n+2} (-1)^{n+1} \|q_{n+1} \alpha\|$

$p_{n+1} = p_n + a_{n+2} p_{n+1}$ $\square$

Q How to compute a_n ?

$$\alpha_{n+1} = \frac{\|q_n \alpha\|}{\|q_{n+1} \alpha\|} = \frac{|\Im_{n-1}|}{|\Im_n|} \quad a_n = \lfloor \alpha_n \rfloor$$

$$\frac{1}{\alpha_{n+1}} = \frac{\|q_n \alpha\|}{\|q_{n+1} \alpha\|} = \frac{\|q_{n-2} \alpha\| - a_n \|q_{n-1} \alpha\|}{\|q_{n-1} \alpha\|} = \alpha_n - \frac{a_n}{\alpha_n}$$

$$\beta_n = \frac{1}{\alpha_n} \quad \beta_{n+1} = \frac{1}{\beta_n} - \left\lfloor \frac{1}{\beta_n} \right\rfloor = \left\{ \frac{1}{\beta_n} \right\}$$

$$\beta_{n+1} = G(\beta_n) \text{ where } G: x \mapsto \left\{ \frac{1}{x} \right\} \text{ Gauss map}$$

$$\alpha_0 = \alpha$$

$$q_0 = 1$$

$$p_0 = 0$$

$$q_1 = \left\lfloor \frac{1}{\alpha} \right\rfloor$$

$$p_1 = 1$$

⚠ If $\alpha \in \mathbb{Q}$ Everything is false at some point the procedure will stop $\alpha = \frac{p}{q} \leadsto q\alpha = 0 \pmod{\mathbb{Z}}$

- Course of C. Uzcaynar made at ICTP 2018?
Video and lecture notes.
- S. Ghazouani, on its web page "circle maps"
Team-mate McRandom